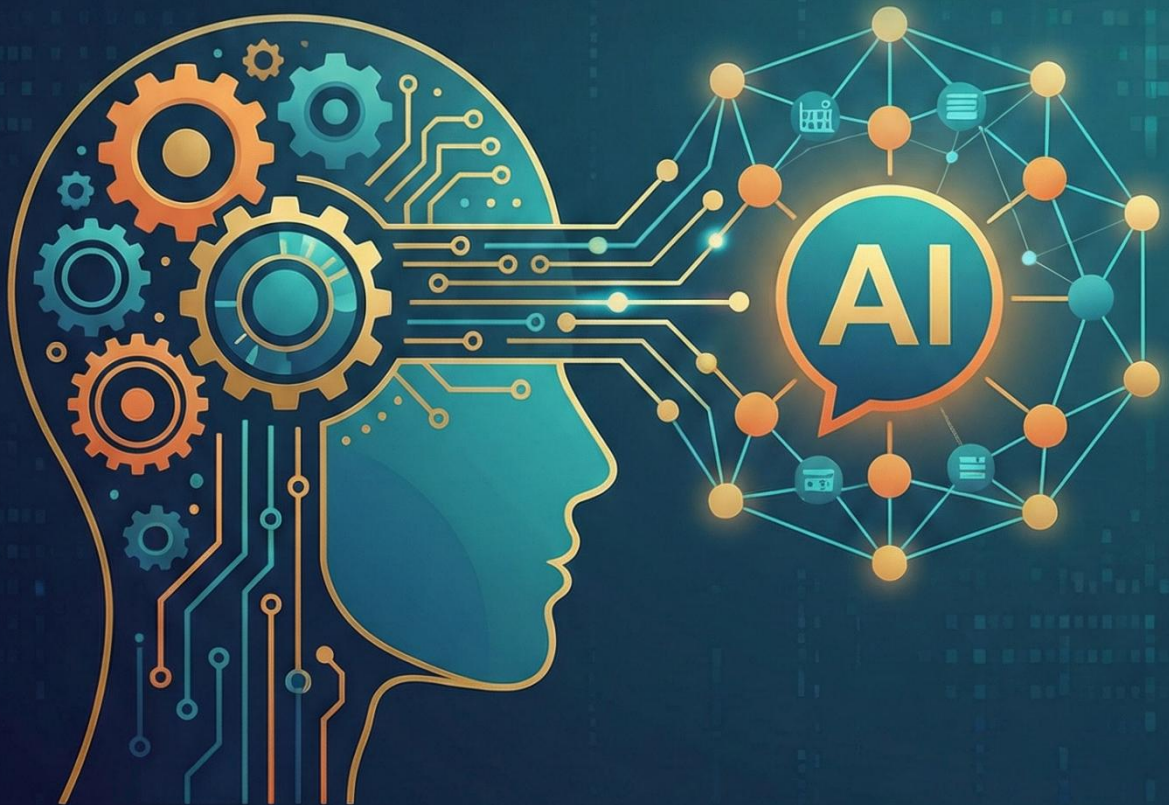


Prompt Engineering Guide

Mastering the Art of Human-AI Communication.
A Comprehensive Resource for Crafting Effective Prompts, Unlocking Creativity and Precision in Artificial Intelligence.



Master Prompt Engineering

Prompt engineering is the art and science of crafting inputs (prompts) that guide Large Language Models (LLMs) to produce the most accurate, relevant, and high-quality outputs possible. As models become more sophisticated, the focus has shifted from "tricking" the model to providing clear, structured context.

Below is a comprehensive guide to the essential rules and techniques of prompt engineering.

1. The Core Components of a Perfect Prompt

A high-quality prompt is rarely a single sentence. It is a structured set of instructions. Most expert prompts include these five elements:

1. **Role/Persona:** Telling the AI who it is (e.g., "Act as a senior data scientist").
2. **Context:** Providing the background or "the why" behind the request.
3. **Task:** A clear, verb-driven instruction (e.g., "Summarize," "Calculate," "Draft").
4. **Constraints:** What the AI *should not* do (e.g., "Do not use jargon," "Limit to 200 words").
5. **Output Format:** Specifying the structure (e.g., Markdown table, JSON, bullet points).

2. Fundamental Rules of Prompting

Rule #1: Be Specific and Explicit

Vague prompts lead to vague results. Instead of saying "Write a story about a cat," say "Write a 500-word suspenseful short story about a stray cat in London who discovers a secret portal in an alleyway." The more parameters you provide, the less the model has to "hallucinate" or guess your intent.

Rule #2: Use Delimiters

Delimiters help the AI distinguish between your instructions and the data you want it to process. Common delimiters include:

- Triple quotes: `"""`
- XML-style tags: `<text></text>`
- Markdown headers: `###`
- Dashes: `---`

Example: `> "Summarize the text delimited by triple backticks. Text: [Insert long article here]"`

Rule #3: Provide Examples (Few-Shot Prompting)

Models are excellent pattern matchers. If you want the AI to output data in a specific, nuanced style, show it three to five examples of what a "good" output looks like. This is known as **Few-Shot Prompting**, whereas providing no examples is **Zero-Shot Prompting**.

Rule #4: The "Chain of Thought" (CoT) Technique

For complex logic, math, or reasoning, don't just ask for the answer. Ask the model to "**Think step-by-step.**" This forces the model to process the logic linearly, which significantly reduces errors in multi-step problems.

3. Advanced Prompting Frameworks

The RTF Framework (Role, Task, Format)

This is the simplest professional framework for quick results:

- **Role:** "You are an expert career coach."
- **Task:** "Review this resume for a software engineering role."
- **Format:** "Provide feedback in a bulleted list categorized by 'Strengths' and 'Improvements'."

The CREATE Framework

For more complex projects, use CREATE:

- **Character:** Define the persona.
 - **Request:** State the specific task.
 - **Examples:** Provide a reference point.
 - **Adjustments:** Add constraints (e.g., "Use a professional tone").
 - **Type of Output:** Specify the file type or layout.
 - **Extras:** Anything else (e.g., "Ask me clarifying questions before starting").
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4. Iterative Refinement and Troubleshooting

Prompting is an iterative process. If the first output isn't perfect, you should "tune" the prompt rather than starting over.

1. E-Commerce Application

Prompt:

Build a full-scale, production-ready E-Commerce application for mobile and web platforms.

The application should allow users to register and log in using email, phone number, or social authentication.

Users should be able to browse products through categories, subcategories, collections, and search with filters such as price, rating, brand, availability, and delivery time.

Each product page should include high-quality images, videos, detailed descriptions, specifications, pricing, discounts, reviews, ratings, FAQs, and delivery estimates.

Implement cart management, wishlist functionality, secure checkout, coupon application, and multiple payment methods including UPI, debit/credit cards, net banking, wallets, and cash on delivery.

Add real-time order tracking, delivery notifications, return and refund workflows, and customer support

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chat.

Create a seller/vendor dashboard where sellers can onboard, upload products, manage inventory, pricing, discounts, and track sales performance through analytics.

Develop an admin panel to manage users, sellers, products, categories, offers, disputes, refunds, and platform analytics.

Integrate AI for personalized product recommendations, smart search, dynamic pricing insights, fraud detection, and demand forecasting.

Ensure the app is fast, scalable, secure, mobile-first, and optimized for high traffic.

2. Healthcare / HealthTech Application

Prompt:

Design a secure, privacy-focused HealthTech application that connects patients, doctors, and healthcare providers.

Patients should be able to create profiles, store medical history, upload lab reports, prescriptions, and track vitals such as blood pressure, glucose levels, heart rate, sleep, and physical activity.

The app should support doctor discovery, appointment booking (online & offline), video consultations, chat consultations, and digital prescriptions.

Doctors should have dashboards to manage schedules, patient records, consultation history, prescriptions, and follow-ups.

Include medication reminders, health alerts, vaccination tracking, and preventive care suggestions.

Integrate AI to analyze symptoms entered by users, provide preliminary health insights, lifestyle recommendations, and risk indicators (non-diagnostic).

Add hospital/clinic dashboards for managing doctors, appointments, and patient flow.

Ensure strict data encryption, role-based access, consent management, and compliance-ready architecture.

Focus on trust, accessibility, multilingual support, and smooth user experience.

3. Education / EdTech Application

Prompt:

Create a comprehensive EdTech learning platform for students, job-seekers, and working professionals.

The platform should support recorded video courses, live classes, assignments, quizzes, projects, certificates, and progress tracking.

Learners should be able to enroll in courses, follow structured learning paths, track completion, and receive reminders.

Instructors should have dashboards to upload content, schedule live sessions, manage learners, evaluate assignments, and analyze engagement metrics.

Include community features such as discussion forums, doubt-solving chat, peer interaction, and mentor support.

Integrate AI to recommend personalized learning paths, summarize lectures, generate quizzes, evaluate answers, and suggest skill improvements.

Add gamification such as badges, leaderboards, and milestones to increase engagement.

Ensure mobile-first design, low-internet optimization, and scalable infrastructure for millions of learners.

FinTech Application

Prompt:

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Build a highly secure, scalable, and regulation-ready FinTech application designed for digital payments, personal finance management, and financial intelligence.

The platform should support **multiple user roles**, including **End Users (Customers), Merchants, Support Agents, and Admins**, each with clearly defined access and dashboards.

1. User Onboarding & Identity Management

The application should allow users to register using mobile number, email ID, and social login options.

Implement OTP-based verification and multi-factor authentication for enhanced security.

Include a complete **KYC workflow**, allowing users to upload identity documents, verify bank accounts, and complete profile verification before accessing financial services.

Provide a clear onboarding flow explaining features, permissions, and security policies in simple language.

2. Account & Wallet Management

Users should be able to create and manage one or multiple digital wallets linked to bank accounts.

Display wallet balance, transaction history, pending transactions, and settlement timelines in real time.

Support adding, removing, and managing multiple bank accounts and payment methods.

Allow users to set spending limits, daily transaction caps, and security preferences.

3. Payments & Transactions

Enable seamless peer-to-peer (P2P) money transfers using UPI, QR codes, phone numbers, or wallet IDs.

Support merchant payments, bill payments, mobile recharges, subscriptions, and recurring payments.

Provide instant transaction confirmations, detailed receipts, and downloadable transaction statements.

Handle failed transactions gracefully with retry mechanisms and transparent error messaging.

4. Expense Tracking & Financial Insights

Automatically categorize transactions into spending categories such as food, travel, shopping, utilities, and savings.

Show daily, weekly, monthly, and yearly spending summaries with visual charts and trends.

Allow users to manually edit or re-categorize expenses if needed.

Provide personalized financial health scores based on income, expenses, savings, and spending habits.

5. Budgeting, Savings & Goal Management

Enable users to create monthly budgets for different categories and track real-time progress.

Allow users to set short-term and long-term financial goals such as emergency funds, travel, education, or

investments.

Show goal progress visually and send smart reminders when users overspend or fall behind savings targets.
Offer automated saving suggestions based on income and spending behavior.

6. Merchant & Business Features

Provide merchant onboarding with business verification and settlement configuration.

Merchants should have dashboards showing daily sales, transaction history, settlements, refunds, and customer analytics.

Support invoice generation, payment links, QR codes, and transaction reconciliation.

Enable customer refund handling and dispute management.

7. AI-Powered Intelligence Layer

Integrate AI to analyze transaction patterns and provide **personalized financial insights**.

AI should detect unusual spending behavior, potential fraud, or abnormal transactions in real time.

Provide smart recommendations such as cost-cutting suggestions, savings optimization, and budget improvements.

Use AI to predict future expenses, upcoming bills, and cash-flow trends.

Ensure AI outputs are explainable, transparent, and user-friendly.

8. Notifications & Alerts

Send real-time notifications for transactions, low balances, upcoming bills, budget breaches, and security alerts.

Allow users to customize notification preferences across SMS, email, and in-app alerts.

9. Customer Support & Dispute Resolution

Include in-app customer support via chat, ticketing, and AI chatbots for common queries.

Enable users to raise disputes for failed or incorrect transactions.

Support agents should have dashboards to track issues, communicate with users, and resolve cases efficiently.

10. Admin & Compliance Management

Admin dashboards should allow monitoring of users, merchants, transactions, risk scores, and system health.

Provide tools for manual review, fraud investigation, user suspension, and audit logging.

Ensure the system is designed with **compliance-ready architecture**, supporting regulatory reporting and data retention policies.

11. Security & Privacy

Implement end-to-end encryption for all sensitive data and transactions.
Use role-based access control, secure APIs, and audit trails.
Ensure strict data privacy, consent management, and secure cloud storage.
Include account lockouts, device binding, and session monitoring.

12. Scalability, Performance & UX

Design the system to handle high transaction volumes with low latency.
Ensure high availability, fault tolerance, and real-time synchronization.
The UI should be clean, intuitive, mobile-first, and accessible to non-technical users.
Optimize for speed, clarity, and trust-building visuals.

13. Future-Ready Enhancements

Design the architecture so that future modules like credit scoring, lending, insurance, investments, or AI agents can be added seamlessly.
Ensure the system can scale across regions, languages, and currencies.

Social Media / Community Application

Prompt:

Build a highly scalable, secure, and engagement-driven Social Media and Community platform designed for content creation, interaction, and meaningful discussions.
The application should support **multiple user roles** including **Users, Creators, Moderators, Community Managers, and Admins**, each with clearly defined permissions and dashboards.

1. User Onboarding & Profile Management

Users should be able to sign up using email, mobile number (OTP-based), or social login options.
The onboarding flow should guide users to select interests, topics, and communities to personalize their experience from day one.
Each user profile should include profile photo, bio, interests, links, followers/following counts, badges, and activity history.
Allow users to manage privacy settings, visibility of posts, comments, followers, and direct messages.

2. Content Creation & Publishing

Users should be able to create and publish different content formats including text posts, images, carousels, short videos, long-form videos, polls, and links.

Provide rich content editing tools such as captions, hashtags, mentions, emojis, tagging, drafts, and scheduling.

Allow creators to save drafts, preview content, and publish across multiple communities where allowed.

Support content editing and deletion with clear version control and transparency.

3. Feed & Discovery System

Design a personalized feed that combines followed content, trending posts, recommended creators, and community highlights.

Include multiple feed views such as “For You”, “Following”, “Trending”, and “Community-Specific Feeds”.

Implement infinite scrolling, real-time refresh, and content prioritization based on relevance and freshness.

Provide advanced search for users, posts, hashtags, communities, and topics.

4. Engagement & Interaction

Users should be able to like, comment, reply, react, share, repost, and save content.

Support threaded comments, pinned comments, and comment moderation by creators.

Include private messaging, group chats, voice notes, and media sharing with read receipts and typing indicators.

Allow users to follow, unfollow, mute, block, and report other users.

5. Community & Group Management

Enable users to create topic-based communities or groups (public, private, invite-only).

Community owners and moderators should manage members, approve posts, enforce rules, and organize content.

Provide tools for community announcements, pinned posts, events, and polls.

Support community analytics such as growth, engagement, and active members.

6. Creator Tools & Monetization

Provide creators with dashboards showing follower growth, reach, impressions, engagement rates, and audience demographics.

Enable monetization features such as subscriptions, paid communities, exclusive content, tipping, and brand

collaborations.

Allow creators to manage pricing, payouts, and subscriber access.

7. AI-Powered Personalization & Intelligence

Integrate AI to personalize content feeds based on user interests, behavior, engagement history, and preferences.

AI should recommend relevant communities, creators, and content.

Implement AI-based content moderation to detect spam, fake accounts, hate speech, misinformation, and harmful content.

Use sentiment analysis to understand community health and engagement quality.

Provide explainable AI recommendations to maintain transparency and trust.

8. Notifications & Real-Time Updates

Send real-time notifications for likes, comments, replies, mentions, messages, community updates, and system alerts.

Allow users to customize notification preferences by type and frequency.

Support push notifications, in-app alerts, and email notifications.

9. Moderation, Safety & Trust

Provide robust reporting tools for users to flag inappropriate content or behavior.

Moderators should have dashboards to review reports, take actions (warn, mute, suspend, ban), and track repeat offenders.

Implement community guidelines enforcement with transparency and appeal mechanisms.

Include user safety features such as content filters, block lists, and parental controls where applicable.

10. Admin & Platform Management

Admins should have full visibility into platform activity, user growth, content trends, and system health.

Provide tools to manage users, creators, communities, monetization, moderation policies, and platform settings.

Include audit logs, analytics dashboards, and manual override controls.

11. Scalability, Performance & Infrastructure

Design the system to support millions of concurrent users with real-time interactions.

Ensure low latency content delivery, caching strategies, and high availability.

Support media optimization, compression, and CDN integration.
Implement failover, monitoring, and performance analytics.

12. Security & Privacy

Implement secure authentication, encrypted messaging, and secure media storage.
Ensure role-based access control and data privacy compliance.
Allow users to download or delete their data as per privacy policies.
Protect against bots, fake accounts, and abuse using AI and behavioral analysis.

13. UX, Accessibility & Inclusivity

Design a clean, intuitive, mobile-first UI optimized for high engagement.
Support dark mode, accessibility features, and multilingual interfaces.
Ensure consistent design language and smooth animations.

14. Future-Ready Architecture

Design the system to easily add future features such as live streaming, AI avatars, VR communities, AI agents, and advanced analytics.
Ensure modular architecture and easy feature expansion.

Travel & Booking Application

Prompt:

Build a comprehensive, scalable, and user-centric Travel & Booking application that helps users **discover, plan, book, and manage travel experiences** seamlessly.

The platform should support multiple user roles including **Travelers (End Users), Travel Service Providers (Hotels, Airlines, Transport, Experience Hosts), Support Agents, and Admins**, each with clearly defined permissions and dashboards.

1. User Onboarding & Profile Management

Users should be able to sign up using email, mobile number (OTP-based), or social login options.
The onboarding flow should collect basic preferences such as travel interests (leisure, business, adventure, budget, luxury), preferred destinations, budget range, language, and currency.
Each user profile should store personal details, saved travelers, passport/ID details (optional and encrypted), payment methods, booking history, wishlists, and loyalty rewards.
Allow users to manage notification preferences, privacy settings, and communication channels.

2. Search, Discovery & Exploration

The application should allow users to search for **flights, hotels, homestays, transportation, and travel experiences** using destination, dates, number of travelers, budget, and preferences.

Provide advanced filters such as price range, ratings, amenities, cancellation policy, duration, stops, airlines, hotel type, and location proximity.

Include interactive maps for location-based discovery.

Show trending destinations, seasonal recommendations, popular itineraries, and curated travel collections.

Support flexible date search and fare calendars.

3. Detailed Listings & Content Pages

Each flight, hotel, or experience listing should include detailed descriptions, images, videos, amenities, inclusions, exclusions, policies, ratings, reviews, FAQs, and host/provider information.

Display transparent pricing with taxes, fees, discounts, and offers clearly highlighted.

Allow users to compare multiple options side-by-side.

Include trust indicators such as verified providers, cleanliness ratings, and safety measures.

4. Booking & Payment Flow

Enable a smooth, multi-step booking process with traveler details, add-ons (meals, luggage, insurance), seat or room selection, and special requests.

Support multiple payment options including UPI, cards, wallets, net banking, EMIs, and international payments.

Provide instant booking confirmation, digital tickets/vouchers, and downloadable invoices.

Handle booking failures gracefully with retries and clear error messaging.

5. Itinerary Management & Trip Dashboard

After booking, users should access a centralized trip dashboard showing all bookings in one place.

Provide day-wise itineraries, travel timelines, check-in/check-out reminders, and location maps.

Allow users to add notes, upload documents, and share itineraries with co-travelers.

Support offline access to tickets and essential trip information.

6. AI-Powered Travel Intelligence

Integrate AI to deliver personalized travel recommendations based on user preferences, past trips, budget, and seasonality.

AI should suggest optimized itineraries, best booking times, alternative routes, and cost-saving options.

Provide AI-generated travel guides, packing lists, and local recommendations.
Use predictive analytics for price forecasting and demand trends.
Ensure AI explanations are clear and user-friendly.

7. Reviews, Ratings & Community Content

Allow users to leave verified reviews, ratings, photos, and tips after completing trips.
Highlight authentic reviews and surface useful insights such as pros, cons, and best times to visit.
Enable Q&A sections for listings and destinations.

8. Customer Support & Issue Resolution

Include in-app customer support via chat, ticketing, and AI chatbots for common queries.
Allow users to modify, cancel, or reschedule bookings based on policies.
Provide transparent refund tracking and resolution timelines.
Support agents should have dashboards to manage queries and escalations.

9. Partner / Provider Dashboards

Travel service providers should be able to onboard, manage listings, pricing, availability, offers, and policies.
Provide analytics on bookings, revenue, customer feedback, and occupancy rates.
Support bulk uploads and integrations with external systems.

10. Notifications & Real-Time Updates

Send real-time notifications for booking confirmations, reminders, delays, cancellations, gate changes, and important travel alerts.
Allow users to customize notification frequency and channels (push, email, SMS).

11. Admin & Platform Management

Admins should have full control over users, providers, bookings, pricing rules, commissions, disputes, and content.
Provide dashboards for platform analytics, revenue tracking, demand insights, and system health.
Include audit logs, manual overrides, and policy enforcement tools.

12. Security, Privacy & Compliance

Implement secure authentication, encrypted payments, and encrypted storage for sensitive user data.
Use role-based access control and secure APIs.
Ensure compliance-ready architecture for data protection and international operations.
Provide clear privacy policies and consent management.

13. Scalability, Performance & Reliability

Design the system to handle high traffic during peak seasons and flash sales.
Ensure low-latency search results, fast checkout, and high availability.
Implement monitoring, alerting, and failover mechanisms.

14. UX, Accessibility & Global Readiness

Design a clean, intuitive, mobile-first user interface with rich visuals and smooth interactions.
Support multiple languages, currencies, and regional formats.
Ensure accessibility features for diverse users.

15. Future-Ready Enhancements

Design the platform to easily integrate future features such as AI travel agents, voice-based search, AR/VR destination previews, loyalty programs, and dynamic packaging.
Ensure modular architecture for continuous expansion.

Food & Grocery Delivery Application

Prompt:

Build a highly scalable, real-time, and reliability-focused Food & Grocery Delivery application that connects **Customers, Restaurants/Grocery Stores, Delivery Partners, Support Agents, and Admins** on a single unified platform.
The application should enable fast ordering, seamless payments, real-time delivery tracking, and intelligent operations optimization.

1. User Onboarding & Profile Management (Customers)

Users should be able to sign up using mobile number (OTP-based), email, or social login.
During onboarding, collect delivery addresses, food preferences (veg/non-veg/vegan), dietary restrictions, allergies, favorite cuisines, and budget preferences.

User profiles should store order history, saved addresses, payment methods, loyalty points, offers, and subscription status (if applicable).

Allow users to manage notification settings, language preferences, and privacy controls.

2. Discovery, Search & Browsing

The app should allow users to discover nearby restaurants and grocery stores based on location and delivery radius.

Provide advanced search and filters such as cuisine type, ratings, price range, delivery time, offers, vegetarian options, and availability.

Show curated sections like “Popular Near You”, “Top Rated”, “Quick Delivery”, “Healthy Options”, and “Daily Essentials”.

Include dynamic menus and real-time inventory visibility for grocery items.

3. Menu & Product Detail Pages

Each restaurant or store page should display images, descriptions, menus or product catalogs, pricing, taxes, delivery fees, preparation time, ratings, reviews, and hygiene/safety badges.

Individual food items or grocery products should show ingredients, nutritional information (if available), portion size, variants, add-ons, substitutions, and availability.

Clearly highlight offers, combos, discounts, and best-selling items.

4. Cart & Checkout Experience

Users should be able to add, remove, and customize items in the cart.

Support multiple carts for different vendors if needed, with clear separation.

Apply coupons, promo codes, wallet balance, and loyalty points at checkout.

Provide a transparent bill breakdown including item cost, taxes, delivery charges, platform fees, and discounts.

Support multiple payment methods including UPI, cards, wallets, net banking, cash on delivery, and subscriptions.

Ensure fast, frictionless checkout with error handling and retry logic.

5. Order Processing & Real-Time Tracking

Once an order is placed, show real-time order status: confirmed → preparing → picked up → out for delivery → delivered.

Display live delivery partner location on the map with estimated delivery time.

Send real-time notifications for order updates, delays, or issues.

Allow users to contact the delivery partner or support if needed.

6. Restaurant / Grocery Store Partner Module

Vendors should onboard through a verification process and create their store profiles.

Provide dashboards to manage menus/products, pricing, inventory, availability, preparation times, and offers.

Vendors should receive incoming orders in real time, update order status, and manage cancellations or substitutions.

Show analytics on orders, revenue, peak hours, customer ratings, and performance metrics.

7. Delivery Partner Module

Delivery partners should onboard with identity and vehicle verification.

Provide delivery partner apps to accept or reject orders, navigate optimized routes, and track earnings.

Show delivery instructions, customer contact options, and order handling guidelines.

Include shift management, incentives, ratings, and payout tracking.

8. AI-Powered Intelligence Layer

Integrate AI to optimize delivery routes, assign orders to the nearest and most efficient delivery partners, and predict delivery times.

Use AI to forecast demand, manage surge pricing, and balance supply during peak hours.

Provide personalized food and grocery recommendations based on user preferences, order history, time of day, and weather.

Detect fraud, fake orders, and abnormal activity.

Ensure AI decisions are explainable and transparent.

9. Subscriptions, Loyalty & Engagement

Support subscription plans for free delivery, exclusive discounts, or priority support.

Implement loyalty programs with points, rewards, and cashback.

Enable gamification such as streaks, challenges, and referral rewards to drive retention.

10. Reviews, Ratings & Feedback

Allow customers to rate and review food quality, delivery experience, and packaging.

Enable photo uploads and structured feedback.

Surface actionable insights to vendors and delivery partners for improvement.

11. Customer Support & Issue Resolution

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Provide in-app support via chat, ticketing, and AI chatbots for common issues.
Allow users to raise complaints for missing items, late deliveries, wrong orders, or refunds.
Support agents should have dashboards to manage tickets, refunds, compensations, and escalations.

12. Admin & Platform Management

Admins should manage users, vendors, delivery partners, orders, commissions, offers, disputes, and platform policies.
Provide dashboards for platform-wide analytics, revenue, growth, order volumes, and service quality.
Include audit logs, fraud monitoring, and manual override tools.

13. Security, Payments & Compliance

Implement secure authentication, encrypted transactions, and secure payment processing.
Use role-based access control for all modules.
Ensure data privacy, consent management, and secure cloud storage.

14. Scalability, Performance & Reliability

Design the system to handle peak-time traffic, flash sales, and festival demand.
Ensure low-latency APIs, real-time synchronization, and high availability.
Implement monitoring, alerts, and auto-scaling infrastructure.

15. UX, Accessibility & Localization

Design a fast, intuitive, mobile-first UI optimized for quick ordering.
Support multiple languages, currencies, and regional preferences.
Include accessibility features and dark mode.

16. Future-Ready Enhancements

Design the platform to easily integrate future features such as AI voice ordering, smart kitchen integration, drone or autonomous delivery, dynamic pricing, and hyper-local supply chain optimization.
Ensure a modular, extensible architecture for long-term scalability.

Entertainment & Media Application

Prompt:

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Build a large-scale, high-performance Entertainment & Media application designed for **video, audio, music, podcasts, and digital content streaming**.

The platform should support **multiple user roles** including **Viewers/Listeners (End Users), Content Creators, Publishers/Studios, Moderators, Support Agents, and Admins**, each with clearly defined permissions, dashboards, and workflows.

1. User Onboarding & Profile Management

Users should be able to sign up using email, mobile number (OTP-based), or social login.

During onboarding, collect preferences such as content type (movies, series, music, podcasts), languages, genres, age rating preferences, and device usage.

User profiles should support multiple sub-profiles (e.g., family profiles, kids profiles) with individual watch history, recommendations, and parental controls.

Store user preferences, viewing/listening history, downloads, playlists, favorites, and subscription status.

Allow users to manage privacy, notifications, autoplay settings, and data usage limits.

2. Content Discovery & Browsing

The app should offer intuitive content discovery through home screens, curated collections, trending lists, editor's picks, and personalized sections.

Provide advanced search with filters for genre, language, release year, popularity, duration, ratings, and content type.

Include category-based browsing such as Movies, Series, Music, Podcasts, Originals, and Live Content (if applicable).

Support smart suggestions and “continue watching/listening” rows.

3. Content Detail Pages

Each content item should have a rich detail page including title, description, trailer/preview, cast/creators, genre, duration, age rating, language options, subtitles, audio quality, and available resolutions.

Display user ratings, reviews, similar content recommendations, and episode lists (for series).

Highlight awards, popularity badges, and editorial highlights where applicable.

4. Streaming & Playback Experience

Provide high-quality streaming with adaptive bitrate, multiple resolutions, and low-latency playback.

Support features like pause/resume, skip, rewind, playback speed control, subtitles, multi-language audio, and screen casting.

Allow offline downloads with expiry controls and device limits.

Ensure seamless playback across devices with sync between sessions.

5. Playlists, Libraries & Personalization

Users should be able to create playlists, add content to favorites or watchlists, and organize libraries. Maintain detailed watch/listen history with resume points. Support smart playlists auto-generated based on mood, genre, or listening habits.

6. AI-Powered Personalization & Recommendations

Integrate AI to analyze user behavior, preferences, watch time, skip patterns, and ratings to generate highly personalized recommendations. AI should dynamically adjust home feeds, suggest new content, and surface hidden gems. Implement AI-based content tagging, similarity detection, and trend analysis. Ensure AI recommendations are explainable and transparent to build user trust.

7. Content Creator & Publisher Module

Content creators and studios should onboard through a verification process. Provide dashboards to upload content, manage metadata, thumbnails, subtitles, release schedules, and monetization settings. Offer analytics on views, watch time, audience demographics, retention, and revenue. Support content scheduling, geo-restrictions, and rights management.

8. Monetization & Subscription Management

Support multiple monetization models including subscriptions, ad-supported content, pay-per-view, rentals, and premium exclusives. Manage pricing plans, free trials, renewals, cancellations, and refunds. Integrate ad insertion for ad-based content with targeting and reporting. Provide creators with transparent revenue dashboards and payout tracking.

9. Social & Community Engagement

Enable users to rate, review, comment, share content, and follow creators. Support watch parties, shared playlists, and social recommendations. Allow moderation of comments and community interactions.

10. Notifications & Engagement Triggers

Send notifications for new releases, upcoming episodes, expiring downloads, personalized recommendations, and subscription updates.

Allow users to customize notification frequency and channels (push, email, in-app).

11. Moderation, Safety & Compliance

Implement AI-based moderation to detect inappropriate content, copyright violations, and abuse.

Support manual moderation workflows, reporting tools, and takedown requests.

Enforce age ratings, parental controls, and regional content compliance.

12. Admin & Platform Operations

Admins should have dashboards to monitor platform performance, content performance, user growth, subscriptions, and revenue.

Provide tools to manage users, creators, content approvals, monetization rules, and moderation policies.

Include audit logs, reporting tools, and manual override capabilities.

13. Performance, Scalability & Infrastructure

Design the system to support millions of concurrent users and high streaming loads.

Use CDN integration, caching, and media optimization for smooth playback.

Ensure high availability, disaster recovery, and real-time monitoring.

14. Security & Privacy

Implement secure authentication, DRM for content protection, encrypted data storage, and secure APIs.

Protect user data, payment information, and content rights.

Provide data access, export, and deletion options as per privacy policies.

15. UX, Accessibility & Global Reach

Design a visually rich, intuitive, and immersive UI with smooth animations.

Support dark mode, accessibility features, multilingual interfaces, and regional customization.

Optimize for mobile, web, smart TVs, and tablets.

16. Future-Ready Enhancements

Design the platform to easily integrate future features such as live streaming, AI-generated content, virtual concerts, AR/VR experiences, voice-based navigation, and AI media agents.
Ensure modular architecture for continuous evolution.

Business & Productivity Application

1. User Onboarding & Account Management

Allow users to sign up via email, mobile OTP, or SSO (Google, Microsoft, etc.).
During onboarding, collect user role, company size, industry, goals (task management, team collaboration, automation, analytics), and preferred tools.
Support individual accounts as well as organization/workspace creation with team invitations.
Enable profile management with role-based permissions, working hours, time zones, and productivity preferences.
Provide data privacy settings, notification preferences, and device management.

2. Workspace, Projects & Task Management

Allow users to create multiple workspaces for different teams or businesses.
Within each workspace, enable project creation with objectives, timelines, owners, and priority levels.
Tasks should support subtasks, dependencies, due dates, assignees, tags, attachments, comments, and status tracking.
Support multiple task views including list view, Kanban board, calendar view, timeline (Gantt), and table view.
Enable recurring tasks, task templates, and bulk actions.

3. Time Management & Productivity Tracking

Provide built-in time tracking for tasks and projects (manual and automatic).
Show daily, weekly, and monthly productivity reports.
Allow users to log work hours, breaks, and deep-focus sessions.
Support goal setting, daily planning, and priority recommendations.
Include focus modes, Pomodoro timers, and distraction tracking.

4. Collaboration & Communication

Enable real-time collaboration through comments, mentions, file sharing, and activity feeds.
Support team chat, project-based discussions, and announcements.
Allow document collaboration with version control and editing history.
Provide meeting notes, agendas, and action item tracking.

5. AI-Powered Productivity Assistant

Integrate an AI assistant that helps users:

- Auto-prioritize tasks based on urgency, workload, and deadlines
- Suggest task breakdowns and timelines
- Generate meeting summaries and action items
- Draft emails, reports, proposals, and documentation
- Recommend workflow optimizations

AI should learn from user behavior, work patterns, and team performance to provide personalized productivity insights.

Ensure AI suggestions are transparent, explainable, and editable by users.

6. Workflow Automation & Integrations

Allow users to build automation rules such as:

- When a task is completed, notify a stakeholder
- When a deadline is missed, escalate to a manager
- Auto-assign tasks based on workload

Integrate with popular tools like email, calendars, cloud storage, CRM, accounting, and communication platforms.

Support API access and webhooks for advanced integrations.

7. Document, File & Knowledge Management

Provide centralized document storage with folders, tags, and permissions.

Enable creation of wikis, SOPs, internal guides, and knowledge bases.

Support search across tasks, files, comments, and documents.

Include versioning, access history, and document approvals.

8. Performance Analytics & Reporting

Provide dashboards showing:

- Task completion rates
- Team productivity trends
- Time utilization
- Project health indicators

Allow custom report creation and data exports.
Use AI to highlight bottlenecks, burnout risks, and under-utilized resources.
Provide predictive analytics for deadlines and capacity planning.

9. HR, Admin & People Management (Optional Module)

Support employee onboarding, role assignments, and access control.
Track attendance, leaves, working hours, and performance metrics.
Enable feedback, reviews, and goal tracking.
Maintain audit logs and activity tracking.

10. Notifications & Engagement

Send smart notifications for task updates, deadlines, mentions, approvals, and automation triggers.
Allow users to customize notification channels and frequency.
Avoid notification overload using AI-based prioritization.

11. Security, Privacy & Compliance

Implement secure authentication, role-based access control, and encrypted data storage.
Support data backups, disaster recovery, and activity logs.
Ensure compliance with organizational data policies and privacy standards.
Allow admins to control data retention and access.

12. Admin & Organization Controls

Admins should manage users, roles, permissions, workspaces, and billing.
Provide tools to configure workflows, templates, automation rules, and integrations.
Include system health monitoring and usage analytics.

13. Performance, Scalability & Reliability

Design the system to support thousands of teams and millions of tasks.
Ensure fast load times, real-time updates, and high availability.
Optimize for mobile, web, and desktop usage.

14. UX, Accessibility & Experience

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Design a clean, minimal, and distraction-free interface.
Support dark mode, keyboard shortcuts, accessibility standards, and responsive layouts.
Ensure smooth onboarding with tutorials and tooltips.

15. Monetization & Pricing

Support freemium, subscription-based, and enterprise pricing models.
Allow feature-based plans, seat-based pricing, and add-on modules.
Provide billing management, invoices, and usage tracking.

16. Future-Ready Enhancements

Ensure modular architecture for future features such as:

- AI agents for autonomous task execution
- Voice-based task creation
- Predictive workload balancing
- Cross-company collaboration
- Advanced analytics and decision intelligence

Build the platform to evolve with the future of work, automation, and AI-driven productivity.

Real Estate Application

Prompt:

Build a **comprehensive, AI-powered Real Estate Application** that connects **buyers, renters, sellers, landlords, brokers, agents, developers, and property managers** on a single platform. The application should support **property discovery, listing management, transactions, communication, analytics, and post-deal services** with a strong focus on **trust, transparency, and user experience**.

The system must support **multiple user roles** including **Property Seekers, Property Owners, Agents/Brokers, Builders/Developers, Legal/Verification Teams, Admins, and Super Admins**, each with role-based dashboards and permissions.

1. User Onboarding & Identity Management

Enable sign-up via email, mobile OTP, and social logins.
During onboarding, capture user intent (buy, rent, sell, lease, invest, manage property).
Allow users to choose roles such as buyer, tenant, owner, agent, or developer.
Create detailed user profiles with verification status, preferences, saved searches, and interaction history.
Support KYC/identity verification for agents, owners, and developers to ensure trust.
Provide privacy controls and communication preferences.

2. Property Listing & Management

Allow property owners, agents, and developers to create detailed property listings.
Listings should include:

- Property type (apartment, villa, plot, commercial, office, shop, land)
- Location (map-based pinning, landmarks, locality insights)
- Price, rent, maintenance, taxes
- Area, layout, floor plans, amenities
- Images, videos, 3D tours, virtual walkthroughs
- Legal status, ownership type, approvals

Support draft listings, approval workflows, and scheduled publishing.
Allow bulk uploads for developers and brokers.
Provide listing performance analytics (views, leads, conversions).

3. Search, Discovery & Personalization

Implement powerful search and filtering:

- Location, budget, size, type, amenities
- Ready-to-move vs under-construction
- Verified properties

Provide map-based search with price heatmaps.
Show AI-driven recommendations based on user behavior, preferences, and budget.
Allow users to save searches, shortlist properties, and receive alerts for matching listings.
Highlight trending, newly added, and high-demand properties.

4. Buyer & Tenant Experience

Enable users to:

- Compare properties side-by-side
- Schedule site visits
- Chat or call agents/owners
- Request documents and details

Provide affordability calculators (EMI, rent affordability, ROI).
Show locality insights such as schools, hospitals, transport, safety, and price trends.
Allow users to submit offers, negotiate prices, and track deal progress.

5. Agent, Broker & Developer Tools

Provide dedicated dashboards for agents and developers.
Enable lead management, follow-ups, reminders, and pipeline tracking.
Show analytics on leads, conversions, revenue, and listing performance.
Support team collaboration for brokerages and real estate firms.
Enable branding, verified badges, and profile promotion.

6. AI-Powered Intelligence & Automation

Use AI to:

- Recommend properties with high match scores
- Predict price trends and investment potential
- Detect fake or duplicate listings
- Auto-tag properties and extract information from images/documents
- Generate property descriptions automatically

Provide AI-driven chat assistants for buyers and sellers.
Offer AI insights on optimal pricing and listing improvements.

7. Communication & Engagement

Support in-app chat, call masking, and appointment scheduling.
Enable automated follow-ups and reminders.
Provide notification system for price drops, new listings, visit confirmations, and deal updates.
Allow feedback and ratings for agents and properties.

8. Transactions, Documentation & Legal Support

Support digital documentation:

- Booking forms
- Agreements
- Verification documents

Enable secure document uploads and sharing.
Integrate legal verification workflows and status tracking.
Support escrow-like payment flows (where applicable).
Provide deal timelines and transparency at every step.

9. Property Management (Post-Deal Module)

Enable landlords and property managers to:

- Manage tenants
- Track rent payments
- Handle maintenance requests
- Send notices and updates

Provide dashboards for property performance and occupancy.
Support service provider integrations for repairs and maintenance.

10. Monetization & Revenue Models

Support multiple monetization options:

- Featured listings
- Subscription plans for agents and developers
- Lead-based pricing
- Advertising
- Transaction commissions

Provide billing dashboards, invoices, and usage tracking.

11. Trust, Safety & Compliance

Implement verified badges and trust scores.
Allow users to report fraud, spam, or misinformation.
Provide moderation tools and audit logs.
Ensure compliance with local real estate regulations and data privacy laws.
Enable transparent dispute resolution processes.

12. Admin & Platform Management

Provide admin dashboards to:

- Approve listings and users
- Monitor platform activity
- Manage monetization and plans
- Handle disputes and escalations

Show system-wide analytics, growth metrics, and operational insights.

13. Performance, Scalability & Infrastructure

Design the system to handle millions of listings and users.
Ensure fast search, low-latency maps, and high availability.
Optimize image and video delivery using CDN.
Support mobile, tablet, and web platforms seamlessly.

14. UX, Accessibility & Experience

Design a clean, trustworthy, and intuitive interface.
Support multilingual users, accessibility standards, and dark mode.
Provide guided onboarding and education for first-time buyers.
Ensure smooth transitions, animations, and minimal friction.

15. Future-Ready Expansion

Build a modular architecture to support:

- AR/VR property tours
- AI investment advisory
- Fractional ownership
- Smart home integrations
- Global real estate expansion



Ensure the platform evolves with future real estate and prop-tech innovations.

Logistics & Supply Chain Application

Prompt:

Build a full-stack, AI-powered Logistics & Supply Chain Management application designed to digitally manage, optimize, and automate end-to-end supply chain operations for manufacturers, distributors, wholesalers, retailers, 3PL providers, fleet operators, warehouse managers, and enterprise businesses. The platform should handle order creation, inventory management, warehousing, transportation, fleet tracking, delivery execution, vendor coordination, and analytics in a single unified system, with support for multiple user roles such as shippers, logistics managers, warehouse staff, drivers, partners, vendors, finance teams, and administrators, each having role-based dashboards and permissions. The application should enable users to onboard via secure authentication, create organizations and warehouses, define supply chain networks, add suppliers, carriers, vehicles, routes, and SKUs, and manage real-time inventory across multiple locations. Users must be able to create and track purchase orders, sales orders, transfer orders, and shipment requests, with full visibility into order status, shipment milestones, delays, and exceptions. The system should support warehouse operations such as inbound receiving, put-away, picking, packing, dispatch, barcode/QR scanning, batch and lot tracking, expiry management, and stock reconciliation. Transportation management features should include route planning, load optimization, vehicle assignment, fuel tracking, driver management, real-time GPS tracking, geofencing, proof of delivery, and automated status updates. The platform should integrate AI and data intelligence to forecast demand, optimize inventory levels, predict

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delivery delays, suggest optimal routes, reduce logistics costs, and identify supply chain risks and bottlenecks. AI should also assist in anomaly detection such as theft, damage, delays, or inventory mismatch. The application should provide real-time dashboards and analytics for inventory turnover, fulfillment rates, on-time delivery, transportation costs, warehouse productivity, and supplier performance. Communication features should allow seamless coordination between shippers, drivers, warehouses, and customers via in-app notifications, alerts, messaging, and automated updates. The system must support document management including invoices, e-way bills, packing lists, bills of lading, and compliance documents with secure storage and access control. Admin and operations teams should have centralized control over users, workflows, pricing rules, SLAs, approvals, and compliance policies. The platform should be highly scalable, secure, and reliable, supporting thousands of shipments, vehicles, and SKUs simultaneously, with real-time synchronization across web and mobile apps. The UI should be intuitive, operationally efficient, and optimized for field usage, with multilingual support and offline capabilities for drivers and warehouse staff. The system should be designed with a modular, future-ready architecture to easily add features like IoT integration, smart sensors, blockchain-based traceability, cross-border logistics, automated warehouses, and autonomous fleet management, ensuring the platform evolves with the future of global supply chains.

AI-First Application

Prompt:

Build a true AI-First Application where artificial intelligence is the core decision-maker, interface, and workflow engine, not just an add-on feature. The application should be designed so that users interact primarily through AI-driven conversations, recommendations, automation, and intelligent actions, while traditional UI components exist only to support and visualize AI outputs. The system should support multiple user roles such as end users, power users, analysts, operators, and administrators, each with AI-personalized experiences and permissions. From onboarding itself, the AI should guide users through goal discovery, context understanding, and preference learning by asking intelligent questions, analyzing inputs, and continuously refining a user intent model. The application must allow users to express goals in natural language rather than fixed forms, enabling them to say what they want to achieve while the AI translates intent into structured workflows, tasks, or system actions. The AI core should be capable of reasoning, planning, memory retention, and multi-step execution, allowing it to break complex goals into smaller actions, decide which tools or modules to invoke, and execute them autonomously or with human approval. The system should maintain short-term and long-term memory of user behavior, preferences, historical actions, and outcomes to continuously improve accuracy and personalization while giving users full control over data visibility and memory reset. The application should integrate real-time data sources, internal databases, external APIs, and tools, allowing the AI to fetch, analyze, and act upon live information. AI should power personalization across the platform by adapting dashboards, recommendations, notifications, and workflows dynamically for each user. The platform must include AI-driven automation where the system proactively suggests actions, executes routine tasks, detects anomalies, predicts outcomes, and prevents failures before they occur. Built-in explainability is critical, meaning the AI should clearly communicate why a recommendation or decision was made and what data influenced it, ensuring transparency and trust. The application should include robust human-in-the-loop controls, allowing users to approve, modify, or reject AI decisions, especially for high-impact actions. Safety, ethics, and governance must be core pillars, with bias detection, content moderation, access control, and policy enforcement embedded directly into the AI workflows. The system should support continuous learning through feedback loops, user corrections, and outcome evaluation, improving performance over time without retraining the entire model from scratch. The UI should be minimal, intuitive, and conversational, supporting text, voice, and multimodal interactions, while also providing visualizations, summaries, and actionable insights generated by the AI. The architecture must be scalable, modular, and future-ready, supporting AI agents, tool orchestration, multi-model strategies, and easy integration of new capabilities as AI technology evolves. The platform should be designed to operate securely at scale, with strong data privacy, encryption, audit

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logs, and compliance controls. Ultimately, the application should feel like an intelligent partner that understands user intent, thinks proactively, takes initiative, and continuously helps users achieve outcomes faster, smarter, and with minimal effort.

Government / Utility Application

Prompt:

Build a secure, scalable, and citizen-centric Government / Utility Application designed to digitally deliver public services, utilities, and governance workflows for citizens, businesses, government departments, field officers, and administrators through a single unified platform. The application should support services such as electricity, water, gas, sanitation, municipal services, public welfare schemes, certificates, permits, grievances, and service requests, while ensuring transparency, accountability, accessibility, and trust. The system should allow citizens to onboard using verified identity mechanisms such as mobile OTP, national ID integration, or government-approved authentication systems, and create unified digital profiles linked to their service history, documents, and benefits. Users should be able to discover services through natural language search, guided workflows, and AI-assisted service discovery rather than complex forms, allowing them to easily apply for connections, pay bills, download certificates, request repairs, register complaints, and track application or service status in real time. The platform must support end-to-end digital workflows including application submission, document upload, verification, approvals, inspections, service execution, and closure, with clear timelines and status visibility at each stage. Utility management features should include billing, metering data ingestion, consumption analytics, tariff management, subsidies, penalties, and automated notifications for due dates, outages, and service updates. Field operations should be supported through mobile interfaces for inspectors and service staff, enabling work orders, geo-tagged updates, photo uploads, digital signatures, and offline functionality in low-connectivity areas. The application should integrate AI to automate document verification, classify grievances, prioritize service requests, predict outages or failures, detect fraud or misuse, and provide multilingual virtual assistants for citizen support. Real-time dashboards and analytics should be available for government officials and administrators to monitor service performance, turnaround times, revenue collection, complaint resolution, resource utilization, and regional demand patterns. Strong governance, security, and compliance must be built into the system, including role-based access control, encryption, audit trails, data retention policies, and compliance with national data protection and digital governance standards. The platform should support inclusivity through multilingual interfaces, accessibility features, voice-based interactions, and simplified flows for elderly and digitally less-experienced users. Admin modules should allow configuration of services, workflows, SLAs, departments, users, and escalation rules without code changes. The architecture should be highly scalable and resilient to support millions of users, high-traffic events, and mission-critical operations with high availability. The system must be designed with a modular, future-ready approach to support integration with smart city systems, IoT-based meters and sensors, GIS mapping, open data portals, inter-department data exchange, and AI-driven policy insights, ensuring the platform evolves with digital governance and smart infrastructure initiatives while maintaining public trust and service excellence.

Gaming Application

Prompt:

Build a high-performance, scalable, and immersive Gaming Application designed to deliver entertainment, engagement, and social interaction across mobile, web, and console platforms. The application should support multiple game types, including casual games, strategy games, role-playing games (RPGs), multiplayer online games, simulation games, puzzle and arcade games, and AI-driven interactive experiences, with modular architecture to easily add new genres. The platform should enable user onboarding via email, social login, and mobile verification, while capturing preferences, skill levels, and

favorite genres to provide personalized recommendations and adaptive difficulty levels. Users should have profiles with achievements, leaderboard rankings, in-game currency, inventory, rewards, and social connections, which can be synced across devices. The application must provide real-time multiplayer support, including matchmaking based on skill level, region, or preferred play style, low-latency networking, and scalable server infrastructure to handle large numbers of concurrent players. Game mechanics should include dynamic levels, challenges, quests, AI opponents, progression systems, in-game purchases, power-ups, and seasonal or time-limited events. Integrate AI-powered game personalization, allowing the system to adapt difficulty, suggest challenges, recommend opponents or co-players, and provide contextual hints based on user behavior. Include social features such as friends lists, chat, voice communication, clans/guilds, tournaments, and leaderboards to increase engagement and retention. The platform should support cross-platform gameplay, cloud saves, and offline modes where possible, ensuring seamless experiences across devices. Monetization features should include in-app purchases, subscriptions, ads, rewards, and virtual economy management, while keeping balance between revenue and player experience. The application must provide analytics dashboards for developers and administrators to track user engagement, session times, churn, in-game transactions, and game performance metrics, including AI-driven insights to optimize gameplay and retention. Security and anti-cheat measures must be integrated to prevent hacks, fraud, and unfair play, including account protection, anti-bot systems, and real-time monitoring. The UI/UX should be visually appealing, intuitive, and responsive, with adaptive graphics, animations, and sound design optimized for both performance and immersion. Accessibility features, localization in multiple languages, and customizable controls should ensure inclusivity. The architecture should be modular, cloud-ready, and future-proof, supporting live updates, expansions, downloadable content (DLCs), augmented reality (AR) and virtual reality (VR) integration, AI-driven NPCs, procedural content generation, and advanced physics or graphics engines. The goal is to build a fully engaging, AI-enhanced, multiplayer-friendly gaming ecosystem that maximizes user retention, monetization opportunities, and long-term scalability.

1. Intelligent Personal Tutor (Education)

Prompt:

Use **Gemini 3.0 + Notebook LM** to create an Intelligent Personal Tutor that acts like a 24/7 interactive teacher for students. The system should:

- Ask the student their subject and level (e.g., “I want to learn algebra, Grade 10”).
- Break down complex topics into step-by-step explanations with examples.
- Generate practice questions and interactive exercises tailored to the student’s pace.
- Dismiss incorrect answers with explanations (“You chose B, but here's why C is correct...”).
- Track student performance over time and identify weak areas.
- Provide summaries after each study session and recommend personalized learning paths.
- Create visual diagrams or code snippets for concepts (e.g., math formulas, science models).

The AI should use Notebook LM for iterative learning activities (quiz engine, personalized exercises, memory of past performances) and Gemini 3.0 for natural language teaching, concept explanation, and context understanding.

2. Smart Career Coach & Resume Generator

Prompt:

Build a Career Coach assistant using **Gemini 3.0 + Notebook LM** that helps users:

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- Identify their career goals based on skills, experience, and interests.
- Generate tailored resumes, cover letters, and LinkedIn summaries.
- Suggest high-impact roles with clear matching scores.
- Provide AI-powered interview preparation with sample questions and responses.
- Track progress through prep sessions and update strengths/weaknesses.
- Provide salary expectations and negotiation tips for roles in India/Global markets.

Use Notebook LM to store user skill profiles, generate drafts, and iterate on resume examples. Use Gemini 3.0 for polished, contextual text generation and conversational guidance.

3. Business Analytics & Recommendation System

Prompt:

Create a Business Intelligence assistant using **Gemini 3.0 + Notebook LM** that takes CSV/excel data input and:

- Cleans and preprocesses the dataset automatically.
- Generates visual dashboards (charts, trend lines, KPI summaries).
- Answers user queries about the dataset in natural language (“What were the top 3 selling products this quarter?”).
- Provides actionable recommendations (“Focus more on product X in region Y”).
- Forecasts trends using historical data.

Notebook LM should host your data cleaning, visualization pipelines, and iterative Q&A loops, while Gemini 3.0 handles natural language interpretation and final insights explanations.

4. AI Tele-Health Assistant

Prompt:

Build a tele-health companion powered by **Gemini 3.0 + Notebook LM** that can:

- Collect patient symptoms via conversation.
- Ask follow-up questions dynamically.
- Provide structured, preliminary health assessments (non-diagnostic).
- Suggest lifestyle recommendations, diet plans, and care tips.
- Schedule doctor consultations and store user medical history.
- Monitor progress and provide reminders.

Gemini 3.0 should handle the empathetic dialogue and medical-context understanding. Notebook LM should store user data, history, and generate clinical summaries and daily care plans.

5. Automated Legal Assistant

Prompt:

Create an AI legal assistant using **Gemini 3.0 + Notebook LM** that can:

- Interpret user questions like “What are my rights under the Indian Consumer Protection Act?”
- Summarize legal documents, contracts, and agreements in simple language.
- Highlight key clauses, risks, and action items.
- Draft legal letters, notices, and templates.
- Generate step-by-step guides for legal processes.

Notebook LM should handle document ingestion, summarization workflows, and draft repositories. Gemini 3.0 should power natural language understanding, explanation, and conversation.

6. E-Commerce Personal Shopping Assistant

Prompt:

Using **Gemini 3.0 + Notebook LM**, build a personal shopping assistant that:

- Understands shopper preferences, past purchases, and budget.
- Suggests products with pros/cons and price comparisons.
- Generates personalized offers and alerts (“Notify me if this drops below ₹1,999”).
- Answers product queries (“Which laptop is best for Python coding under ₹50,000?”).
- Tracks order history and suggests reorders.

Gemini 3.0 should handle conversational queries and recommendation logic. Notebook LM should store user preferences, build product scoring models, and manage context over time.

7. AI-Driven Study Notes & Concept Summarizer

Prompt:

Build a study assistant using **Gemini 3.0 + Notebook LM** that:

- Takes long passages (PDFs, textbooks, lecture transcripts).
- Generates concise, structured summaries for key concepts.
- Produces visual mind maps or flashcards.
- Creates practice questions based on the summaries.
- Tracks learning retention via spaced repetition reminders.

Gemini 3.0 should handle comprehension, summarization, and conversation. Notebook LM should index note history, generate flashcards, and schedule spaced revision plans.

8. AI Executive Assistant

Prompt:

Create an AI executive assistant using **Gemini 3.0 + Notebook LM** that can:

- Read and summarize emails.
- Draft meeting agendas, minutes, and follow-ups.
- Schedule calendars intelligently (“Move my 3 PM if today’s board meeting runs late”).
- Generate performance reports from data (sales, churn, user growth).
- Provide reminders and priority flags for tasks.

Gemini 3.0 should manage dialogue, summarization, and intent interpretation. Notebook LM should track context, drafts, and historical decisions.

9. Automated Data Pipeline Builder

Prompt:

Use **Gemini 3.0 + Notebook LM** to build a self-serving data pipeline assistant that:

- Ingests data from APIs, CSVs, databases.
- Cleans and transforms data.
- Generates ETL scripts and reusable workflows.
- Builds and schedules dashboards, tables, and data alerts.
- Answers analytical questions based on the pipeline outputs.

Gemini 3.0 should provide natural language interface and logic planning. Notebook LM should manage data workflows, scripts, scheduling, and outputs.

10. AI Creativity Studio (Content & Media)

Prompt:

Build an AI Creativity Studio using **Gemini 3.0 + Notebook LM** that supports:

- Script writing for videos and podcasts.
- Storyboarding, scene breakdowns, and shot lists.
- Auto-editing suggestions and narrative polishing.
- Image/graphic generation with design prompts.
- Multi-track audio editing and automatic mix suggestions.

Gemini 3.0 should power natural language ideation, structure, and refinement. Notebook LM should orchestrate versioning, editing pipelines, asset management, and iterative creative interaction.

11. AI Code Assistant & Learning Mentor

Prompt:

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Use **Gemini 3.0 + Notebook LM** to build an AI developer assistant that helps users learn coding and debug code in real time. Gemini should interpret coding questions, generate clean code in languages like Python/JavaScript/SQL, and explain logic step-by-step. Notebook should run test cases, suggest improvements, handle versioning of code snippets, and generate project templates based on user goals like “Build an e-commerce API” or “Create a recommendation engine”. Provide interactive feedback loops so users can iterate on code and receive explanations for errors.

12. Intelligent Inventory Planning Assistant

Prompt:

Build an AI tool using **Gemini 3.0 + Notebook LM** for inventory forecasting and demand optimization. Gemini should interpret business questions like “How much stock do we need for next month?” and Notebook should analyze historical sales data to forecast demand, suggest re-order quantities, and generate replenishment schedules. Provide insights such as safety stock levels, supplier lead times, and real-time alerts for low stock across multiple warehouses.

13. Virtual Finance & Investment Advisor

Prompt:

Create a virtual investment advisor where users can ask natural language questions like “What’s the best mutual fund to invest in for 5-year growth?” or “Analyze my portfolio risk”. Gemini should explain financial concepts, risk profiles, and goals. Notebook should process user holdings, run risk simulations, asset allocation models, and generate personalized investment plans including projected returns, diversification suggestions, and tax implications.

14. Smart Travel Planner

Prompt:

Build a travel planning assistant using **Gemini 3.0 + Notebook LM** where users can enter broad goals like “I want a 7-day Japan itinerary with local food, budget \$2000”. Gemini should ask follow-ups, refine preferences (culture, pace, attractions). Notebook should generate day-by-day itinerary, booking links, transport suggestions, costs, maps, and weather forecasts. Include packing lists, reservation timelines, and localized tips.

15. Personalized Health & Wellness Coach

Prompt:

Develop an AI wellness coach that guides users through fitness, diet, and mental wellness. Gemini should conversationally understand goals (“I want to lose 5 kg in 3 months with knee-friendly workouts”). Notebook should create day-by-day workout plans, meal calendars, progression dashboards, and track progress over time, adjusting recommendations based on user feedback and biometrics tracking.

16. AI-Enhanced Market Research Assistant

Prompt:

Build a market research tool using **Gemini 3.0 + Notebook LM** that can ingest surveys, web scraped data, and industry reports. Gemini should interpret research objectives and generate summaries. Notebook should analyze sentiment, competitive landscapes, trends, and output charts, SWOT analysis, and executive summaries. Provide exportable reports for investors or product teams.

17. Automated Compliance & Policy Checker

Prompt:

Create a compliance assistant where users can upload policy documents, regulations (e.g., GDPR, PCI-DSS). Gemini should interpret regulatory requirements conversationally. Notebook should parse documents, check rules against company policies, highlight violations, and recommend remediation steps with risk severity levels.

18. Smart Contract Drafting & Review Assistant

Prompt:

Build an AI assistant for contract drafting and review. Gemini should interpret requests like “Draft a freelance agreement for mobile app development”. Notebook should generate structured contract templates, highlight terms, flag legal risks, and compare clauses against best-practice libraries. Provide logical reasoning behind suggested changes.

19. Research Paper Analyzer & Citation Assistant

Prompt:

Create a research assistant where academics upload papers/PDFs. Gemini should summarize key contributions, methods, and results. Notebook should extract tables, figures, citations, generate related work summaries, and produce formatted citation lists in styles like APA or IEEE. Provide idea suggestions and potential future work directions.

20. Social Listening & Brand Sentiment Tracker

Prompt:

Build a marketing intelligence tool that ingests social media feeds and reviews. Gemini should interpret high-level questions like “How is our brand perceived this quarter versus last?”. Notebook should run sentiment analysis, topic extraction, trend charts, competitive brand comparisons, and alert triggers for spikes in negative sentiment.

21. Personalized Real Estate Investment Analyzer

Prompt:

Use **Gemini 3.0 + Notebook LM** to build a property investment advisor. Gemini should interactively understand user goals (“Invest ₹50 L–₹1 Cr in Tier-2 cities for rent yield”). Notebook should analyze price trends, rental yields, crime data, amenities, ROI projections, and generate side-by-side investment reports with scorecards.

22. AI Operations (AIOps) Assistant for IT Teams

Prompt:

Build an IT operations assistant where Gemini helps interpret alerts, incidents, and natural language issues (“Server X’s CPU spikes at night”). Notebook should analyze logs, correlate incidents, suggest root causes, propose mitigation steps, and create runbooks. Provide automated SLA tracking and alert escalation workflows.

23. AI Internal Knowledge Base & Q&A Bot

Prompt:

Create a knowledge assistant that ingests internal documents, SOPs, and manuals. Gemini should answer employee queries like “How do I file my expense report?” conversationally. Notebook should index content, extract relevant sections, provide context-aware answers, and link to source documents. Include content update pipelines.

24. Customer Support Augmentation Tool

Prompt:

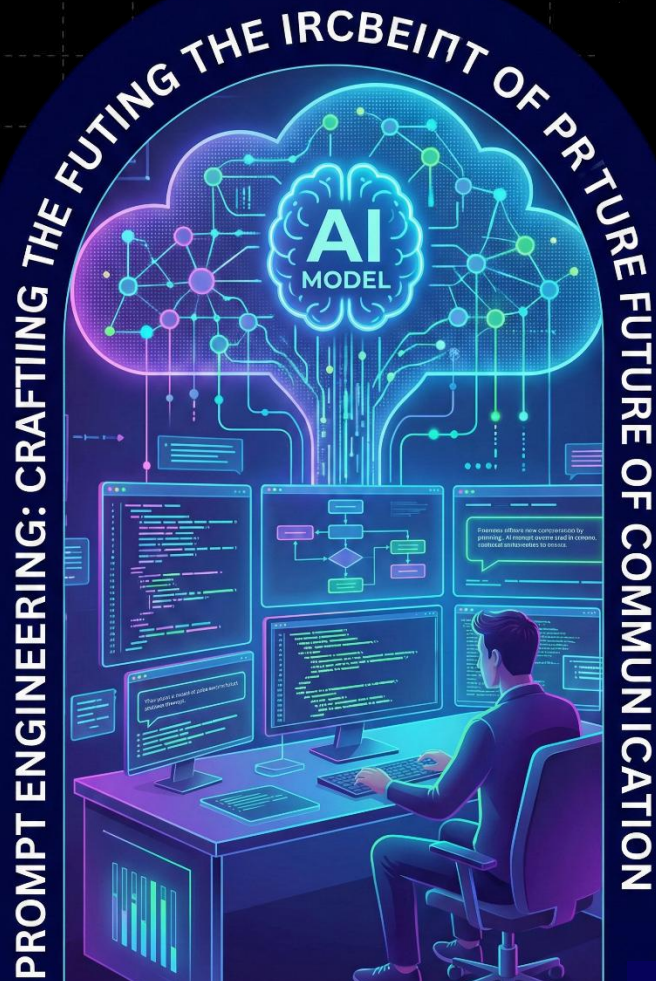
Build an AI support assistant that reads support tickets and provides recommended replies. Gemini should understand sentiment and intent. Notebook should route tickets, auto-draft replies, suggest escalation levels, track response performance, and summarize trends week-over-week.

25. Culinary Creativity & Recipe Engine

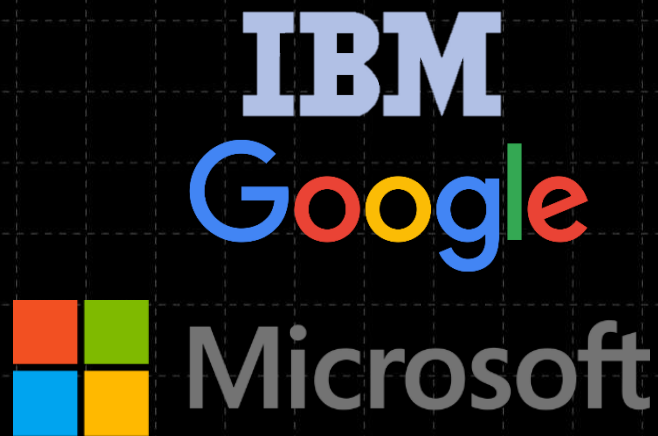
Prompt:

Build a recipe generator where users input ingredients they have at home. Gemini should ask flavor preferences (“spicy, mild, vegan?”). Notebook should generate customized recipes, shopping lists, step-by-step instructions, nutritional breakdown, and cooking timers.



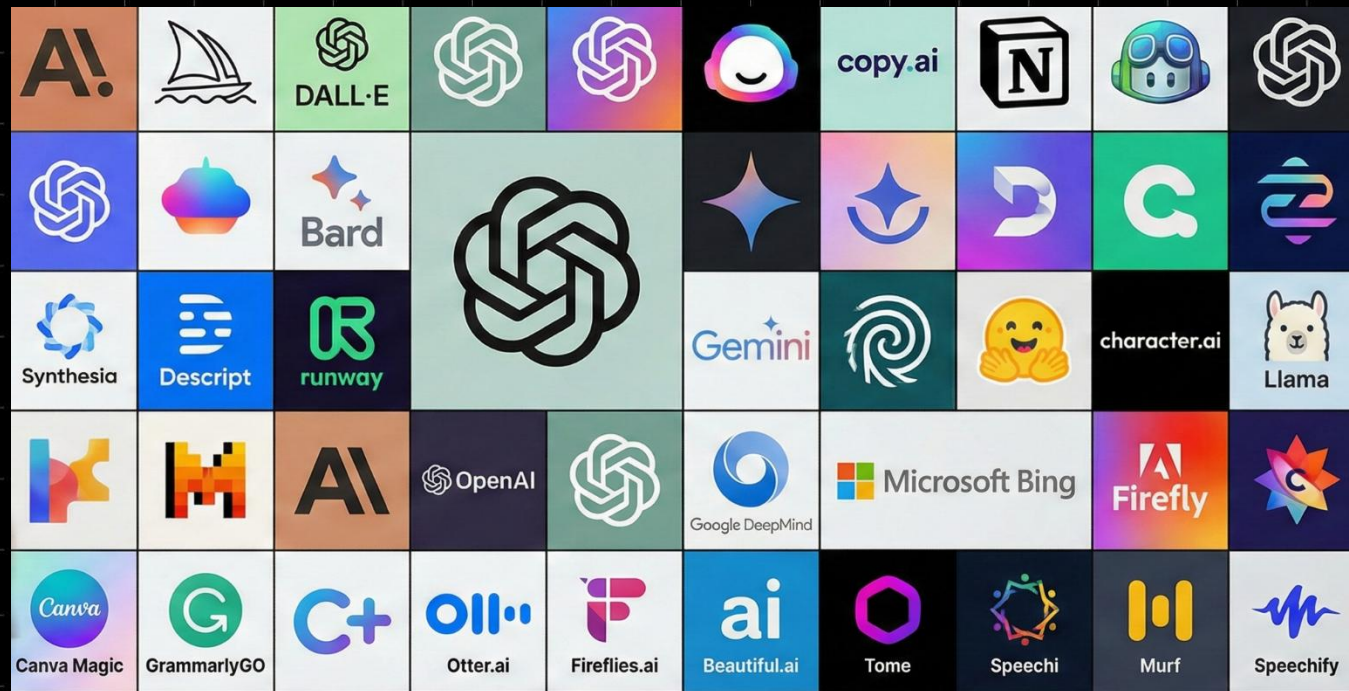


Prompt Engineering Full Course



Prompt Engineering

Full Course with Practical Implementation



Topics to be Covered

- Introduction to LLMs & Prompting Basics
- Evolution: Generative AI vs. Discriminative AI
- How LLMs Work (The Technical Foundation)
- The Transformer Architecture
- Tokens and Context Window
- What is Prompt Engineering
- GIGO Principle
- Anatomy of a Perfect Prompt
- Types of Prompts
- RNN (Recurrent Neural Networks)
- LSTM (Long Short-Term Memory)
- Context Window
- Direct Prompting
- Structured Prompting
- Hallucinations
- Use Case 1: Writing & Content Creation
- Use Case 2: Coding & Technical Tasks
- Use Case 3: Data Analysis & Business Intelligence
- Zero-Shot Prompting
- Few-Shot Prompting
- Chain of Thought (CoT)
- Instruction Prompting
- Technical Aspects & API Parameters
- ReAct Framework
- Tree of Thoughts (ToT)
- Directional Stimulus Prompting
- Iterative Prompt Development
- Data Extraction: Unstructured to Structured
- Prompting for Coding:
- RAG (Retrieval-Augmented Generation)
- Automating Workflows
- Practical Use Cases on AI
- The Dynamic Mock Interviewer
- The Legal/Contract "Risk Finder"
- The Business Strategy Consultant

Module 1 - Introduction to LLMs & Prompting Basics

Evolution: Generative AI vs. Discriminative AI

Discriminative AI

Discriminative AI ka kaam hota hai Data ko classify karna.

Real-world Example

- Email - Spam & Not Spam
- Face ID
- Netflix Recommendation

Generative AI

Generative AI sirf categories mein divide nahi karti, balki Naya Data Generate karti hai jo pehle kabhi exist hi nahi karta tha

Real World Example

- ChatGPT (Text generation)
- Midjourney/DALL-E (Image generation)
- GitHub Copilot (Code generation)

Feature	Discriminative AI	Generative AI
Main Goal	Classification	Creation
Prompt Engineering	Zero Role	Most Important Role

Module 1 - Introduction to LLMs & Prompting Basics

Evolution : RNN, LSTM, Transformer

Phase 1: RNN (Recurrent Neural Networks)

Problem: Inki "memory" bahut kamzor thi. Agar ek bada paragraph hai, toh RNN paragraph ke khatam hone tak yeh bhool jata tha ki shuruat mein kya likha tha. Isse "Vanishing Gradient Problem" kehte hain.

Phase 2: LSTM (Long Short-Term Memory)

Concept: RNN ka upgrade. Isme "Gates" lagaye gaye jo AI ko yeh decide karne mein madad karte the ki kaunsi information "yaad rakhni" hai aur kaunsi "bhoolni" hai.

Problem: Yeh sequential kaam karte the (ek ke baad ek word). Is wajah se inko train karna bahut slow tha aur yeh bahut lambe contexts ko handle nahi kar paate.

Phase 3: Transformers (The Game Changer - 2017)

Concept: Google ne ek paper nikala "Attention is All You Need". Transformers ne "Attention Mechanism" introduce kiya.

Magic: Yeh poore sentence ko ek saath (parallelly) process karte hain, na ki ek-ek word karke. Yeh samajh sakte hain ki sentence ke shuruat ka word, aakhri word se kaise juda hai.

Result: Isi technology par GPT (Generative Pre-trained Transformer) bana hai.

Prompt Engineering

Transformers ki wajah se hi hum Prompt Engineering kar pa rahe hain, kyunki:

1. **Long Context:** Transformers lambe prompts ko yaad rakh sakte hain (kyunki woh sequential nahi, parallel hain).
2. **Contextual Understanding:** Aap ek hi word ko alag-alag context mein use karein, AI samajh jayega (Jaise 'Bank' wala example).
3. **Few-Shot Learning:** In models ne itna zyada data dekha hai ki aap sirf 2-3 examples (shots) dein, toh yeh naya task seekh jaate hain.



What is Prompt Engineering?

Definition: Yeh ek aisi discipline hai jisme hum LLMs (Large Language Models) ke liye aise inputs (Prompts) design karte hain jo model ko guide karein taaki woh sabse zyada Accurate, Reliable, aur Useful output de sake.

AI systems cannot read long sentences the way humans do.

They require text to be converted into small, structured pieces → tokens → which can later be converted into numbers.

Token kya hota hai?

Token ka simple matlab:

- ✓ Words ka small piece
- ✓ Ek pura word
- ✓ Ek character
- ✓ Ya word ke parts (subwords)



AI Words nahi, Tokens padhta hai

Rule: 1 token hamesha 1 word nahi hota. Ek lamba word (jaise: "Unbelievable") 2-3 tokens mein toot sakta hai.

Tokenization Process

- Word: **"Playing"** → AI ise **"Play"** aur **"ing"** do tokens mein dekh sakta hai.
- Word: **"Transformation"** → Ise **"Trans"**, **"form"**, aur **"ation"** mein toda ja sakta hai.

Kyun zaroori hai? Isse AI ko prefixes (re-, un-) aur suffixes (-ing, -ed) samajhne mein madad milti hai, jisse woh naye words ka matlab nikal pata hai.



ChatGPT



Midjourney

Gemini



Jasper



Copilot

Tokens

AI Tool / Model	Approx. Context Window / Token Limit
◆ ChatGPT (OpenAI)	~128,000–256,000 tokens (depending on version)
◆ Google Gemini	~1,000,000–2,000,000 tokens
◆ Anthropic Claude	~200,000 tokens (standard) / Up to ~1,000,000 tokens (enterprise)
◆ Perplexity AI (Sonar/Backends)	~128,000–200,000 tokens (native LLM limits)

Context Window - AI ki "Memory Limit"

Blackboard Analogy

Aap jo bhi prompt likhte hain aur AI jo bhi answer deta hai, woh us blackboard par likha jata hai.

Jab blackboard bhar jata hai, toh naya likhne ke liye AI ko sabse upar (shuruat) ki lines mitani padti hain.

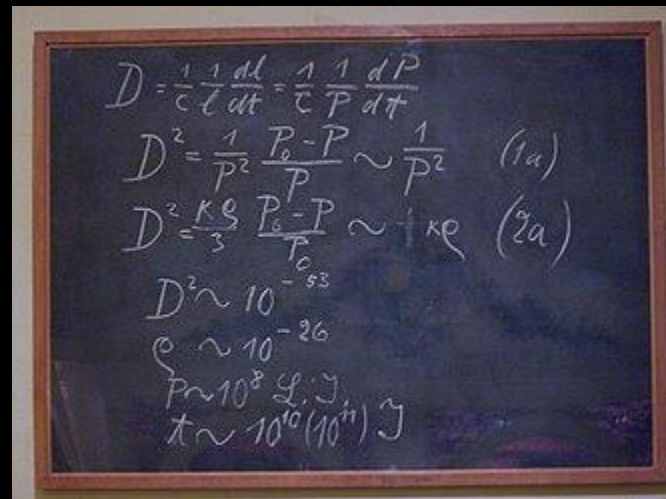
Technical Definition: Ek AI model ek baar mein jitne maximum tokens "dekh" ya "yaad" rakh sakta hai, use Context Window kehte hain.

2. Limit ka Asar

Agar aapka model 8,000 tokens ki limit rakhta hai aur aapne 10,000 tokens ka document de diya:

AI shuruat ke 2,000 tokens ko "bhool" jayega.

Isliye, agar aapne shuruat mein koi instruction di thi (e.g., "Always reply in Hindi"), toh AI baad mein English mein baat karne lagega kyunki woh instruction blackboard se mit chuki hai.



Context Window - AI ki "Memory Limit"

ChatGPT

Excellent for long discussions, essays, emails, summarization within its ~128K–256K window.

Gemini

Among industry-largest context windows available commercially (1M–2M tokens).

Great for:

- Processing multiple book-length docs at once

- Large codebase analysis in one snapshot

- Complex project tasks across long chats.

Claude

Starts at ~200K tokens for most users, ideal for large document reasoning with more safety and ethical guardrails

Perplexity AI

Uses various backend models but typically up to ~200K tokens in Sonar Pro; can behave like shorter context models in basic chats.

Types of Prompts: Direct vs. Structured Prompting

1. Direct Prompting

Ise hum Instruction Prompting bhi kehte hain. Isme hum AI ko ek seedha task dete hain bina kisi extra context ya detail ke.

Characteristics: Short, one-line, aur bahut general hota hai.

Best for: Bahut simple tasks ke liye (e.g., "What is the capital of India?").

Problem: Isme AI ko "guess" karna padta hai ki aapko kya chahiye, jis wajah se output hamesha perfect nahi hota.

Example: > "Write a blog post about coffee." (AI ko nahi pata ki blog kitna lamba ho, kis audience ke liye ho, ya uska tone kaisa ho.)

Types of Prompts: Direct vs. Structured Prompting

Structured Prompting

Role (Persona): AI ko ek "Identity" dena.

Example: "Act as a Senior Content Marketer with 10 years of experience."

Context (Background): Batana ki yeh kaam kyun ho raha hai.

Example: "I am launching a new organic coffee brand for health-conscious Gen-Z."

Task (Action): Clear instruction dena.

Example: "Write an engaging blog post title and a 300-word intro."

Constraints (Negative Constraints): Batana ki kya nahi karna hai.

Example: "Avoid using jargon like 'best-in-class'. Don't mention caffeine side effects."

Format (Output Structure): Batana ki result kaisa dikhna chahiye.

Example: "Output should be in Markdown format with a catchy H1 header."

Use Case 1: Writing & Content Creation

Simple Prompt: "Write a cold email for a Google internship." (General response milega).

Engineered Prompt: "Act as a Career Coach. Write a personalized cold email to a Recruiter at Google for a Data Science Internship. Mention my proficiency in Python and SQL. Keep the tone professional but eager, and limit it to 150 words."

Use Case 2: Coding & Technical Tasks

Simple Prompt: "Fix this code: [code block]."

Engineered Prompt: "Act as a Senior Python Developer. Review the following code for a 'Log-in Logic'. Identify potential security vulnerabilities and logical errors. Provide the corrected code with comments explaining each fix."

Use Case 3: Data Analysis & Business Intelligence

Simple Prompt: "Analyze this sales data."

Engineered Prompt: "I will provide a list of sales data for the last 6 months. Act as a Data Analyst. Identify the top 3 performing products, the month with the highest growth, and suggest 2 strategies to improve sales for the lowest-performing category. Present the final answer in a Table format."

Module 2 - Core Prompting Techniques

1. Zero-Shot Prompting: The "Direct Order"

Zero-Shot ka matlab hai ki aapne AI ko koi Example (Shot) nahi diya. Aapne bas ek instruction di aur ummeed ki ki AI apne pre-trained knowledge ke basis par answer de dega.

Practical Use Case (Sentiment Analysis):

Prompt: "Classify the sentiment of this review: 'The battery life of this laptop is amazing but the screen is a bit dull.'"

Output: "Sentiment: Mixed/Neutral"

Module 2 - Core Prompting Techniques

2. Few-Shot Prompting: The "Pattern Maker"

Kabhi-kabhi Zero-Shot fail ho jata hai, khaas karke jab aapko output ek Specific Format mein chahiye ho. Tab hum use karte hain Few-Shot Prompting. Isme hum AI ko 2-3 examples (shots) dete hain taaki woh pattern samajh jaye.

Practical Use Case (Formatting Data): Aapko customer feedback ko ek specific JSON format mein chahiye.

Prompt: Example 1: "I loved the pizza!" -> {"sentiment": "positive", "label": 1} Example 2: "The delivery was late." -> {"sentiment": "negative", "label": 0} Example 3: "The food was okay." ->

Output: {"sentiment": "neutral", "label": 0.5}

Module 2 - Core Prompting Techniques

3. Chain of Thought (CoT): The "Logical Thinker"

Yeh sabse important technique hai complex problems solve karne ke liye. Jab hum AI ko bolte hain "Think step-by-step", toh woh seedha answer par koodne ki jagah logic banata hai. Isse "Hallucinations" (AI ka jhooth bolna) kam ho jati hain.

show emoji of seahorse

Kyun zaroori hai: Complex math, coding logic, ya strategy banane mein AI aksar jaldrazi mein galat answer deta hai. CoT use "sone (reasoning)" par majboor karta hai.

Practical Use Case (Math/Logic Problem):

Prompt: "A farmer has 15 sheep. All but 8 die. How many are left? Let's think step by step." AI ka Logic: > 1. Total sheep = 15. 2. "All but 8 die" ka matlab hai ki sirf 8 bache hain aur baaki mar gaye. 3. Isiye, jo bache hain unki sankhya 8 hai.

Final Answer: 8

4. Instruction Prompting: The "Framework"

Instruction prompting mein hum AI ko sirf task nahi dete, balki use ek Boundaries (Deewar) mein bandhte hain. Isme hum teen cheezein use karte hain:

Persona (Role): "Act as a Senior Python Developer."

Constraints (Limit): "Don't use external libraries. Keep code under 20 lines."

Specific Instruction: "Explain the code like I am 5 years old."

Practical Use Case (Professional Email):

Prompt: "Act as a Polite HR Manager. Write a rejection email for a candidate who cleared the interview but we can't hire due to budget cuts. Mention that we will keep their CV for future. Constraint: Keep it empathetic and don't mention the exact budget amount."

Module 3 –Advanced Prompt Engineering Frameworks

1. ReAct Framework (Reason + Act)

ReAct ka matlab hai Reasoning + Acting. Pehle AI sirf "sochta" tha ya sirf "answer" deta tha. ReAct mein AI ek loop mein kaam karta hai: woh pehle ek Thought (soch) banata hai, phir ek Action (kaam) karta hai (jaise Google search karna), aur phir Observation (nateeja) dekh kar agla kadam uthata hai.

Use Case: Researching a market trend using AI with browsing capabilities (like ChatGPT with Search or Perplexity).

Prompt: "Solve the following task using a Thought-Action-Observation loop. Task: Analyze the current stock performance of NVIDIA compared to its competitors in the last 7 days. Instructions: > 1. Start with a 'Thought' (What do you need to find?). 2. Perform an 'Action' (Search for specific data). 3. Provide an 'Observation' (Summarize what you found). 4. Repeat until you can provide a final 'Recommendation' for a short-term investor."

Tree of Thoughts (ToT)

Tree of Thoughts "Chain of Thought" ka bada bhai hai. CoT mein AI ek hi line mein sochta hai. ToT mein AI ek Ped (Tree) ki tarah alag-alag raste (branches) explore karta hai.

Kaise kaam karta hai: AI ek saath 3-4 alag ideas (thoughts) sochta hai. Phir woh khud hi evaluate karta hai ki kaunsa idea sabse best hai. Jo rasta bekar lagta hai, use wahin chhod deta hai aur best raste par aage badhta hai.

ToT: AI Strategy A, B aur C sochega. Phir sochega, "Strategy B mein paisa zyada lag raha hai, ise cancel karo. Strategy A sabse sasti aur fast hai, is par aage badhte hain."

Prompt: "Imagine three different experts are brainstorming a solution to this problem: 'How can a small local bakery increase its sales by 50% in 3 months with a budget of only \$500?'

Step 1: Each expert will propose one unique 'thought' or strategy. Step 2: They will critique each other's ideas, identifying risks and costs. Step 3: Based on the critique, they will discard the weakest ideas and combine the best elements into a single, optimized master plan.

Give me the final optimized master plan."

Directional Stimulus Prompting (Hint-based Prompting)

Kaise kaam karta hai: Aap main instruction dete hain aur saath mein kuch "Keywords" ya "Direction" dete hain jo AI ko focus karne mein madad karte hain.

Practical Use Case: Summarization (Saar nikalna).

Prompt: "Is article ko summarize karo."

Stimulus: "Focus on: Financial impact and Future growth."

Result: AI poore article mein se sirf unhi baaton par focus karega jo aapne stimulus mein di hain.

Prompt: `""Summarize the following article about Renewable Energy.`

Stimulus/Keywords to focus on: 'Government Subsidies', 'Solar Panel Efficiency', and 'Employment Opportunities'.

Article: [Paste your text here]"

Iterative Prompt Development (The Improvement Cycle)

Yeh koi technique nahi, balki ek Systematic Process hai. Ek "Prompt Engineer" kabhi bhi pehli baar mein perfect prompt nahi likhta. Woh ek cycle follow karta hai:

Draft: Pehla prompt likho.

Test: Use AI par chalao.

Analyze: Dekho result mein kya kami hai (Kya yeh bahut lamba hai? Kya tone galat hai?).

Refine: Prompt mein badlav karo.

Repeat: Tab tak karo jab tak perfect result na mil jaye.

Iterative Prompt Development (The Improvement Cycle)

Use Case: Building a high-quality "Resume Reviewer" tool.

Iteration 1 (Vague):

"Review this resume and tell me if it's good: [Resume Text]"

Iteration 2 (Adding Persona & Context):

"Act as a Senior HR Manager at a Top-tier Tech Company. Review this resume for a 'Junior Software Engineer' role. Identify 3 strengths and 3 weaknesses."

Iteration 3 (The Final Optimized Version):

"Act as a Senior Tech Recruiter at Google. Task: Conduct a detailed audit of the provided resume. Evaluation Criteria: ATS-friendliness, Quantifiable Achievements (e.g., 'increased sales by X%'), and Keyword Relevance for Python/Cloud roles. Output Format: > 1. ATS Score (0-100). 2. Critical Missing Keywords. 3. Suggestions for rewriting the 'Experience' section to be more impactful. Resume: [Resume Text]"

Framework	Best English Trigger Word/Phrase	Goal
ReAct	"Thought, Action, Observation loop"	Logical Execution / Research
Tree of Thoughts	"Imagine 3 experts brainstorming..."	Comparing multiple solutions
Directional Stimulus	"Focus specifically on these keywords..."	Guided/Refined Output
Iterative Dev	"Refine the persona and constraints"	Reaching Production Quality

Module 4 – Data & Automation for Professionals

1. Data Extraction: Unstructured to Structured

Concept: Aksar professionals ke paas data messy format mein hota hai (emails, meeting notes, receipts). Inhe Excel (CSV) ya Programming (JSON) format mein badalna bahut mehnat ka kaam hai. Prompt Engineering ise seconds mein kar sakti hai.

Real-Life Example: Ek HR manager ke paas 50 candidates ka intro text hai aur unhe ek Excel sheet banani hai jisme Name, Experience, aur Skills alag-alag hon.

Practical English Prompt:

"Act as a Data Processing Expert. Extract the following information from the provided text: Name, Years of Experience, Current Role, and Primary Skills. Format the output as a valid JSON array of objects. Text: [Paste Candidate Bios here]"

Module 4 – Data & Automation for Professionals

2. Prompting for Coding: The Developer's Assistant

Code Refactoring (Code ko Saaf aur Fast Banana)
Refactoring ka matlab hai code ka "Kaam" (logic) wahi rahe, lekin uski "Quality" behtar ho jaye.

Prompt Example: "I have a working JavaScript function that uses multiple nested if-else statements. It is hard to read. Refactor this code using a switch statement or a 'Lookup Object' to make it cleaner and more efficient. Maintain the same functionality. Code: [Paste your code here]"

Module 4 – Data & Automation for Professionals

RAG (Retrieval-Augmented Generation): AI with its own Library

Concept: LLMs (ChatGPT/Claude) ko sirf wahi pata hai jis par woh train huye hain. Agar aap unse apne "Company ki internal policy" ya "Kal ki news" ke baare mein puchenge, toh woh jhooth bol sakte hain. RAG AI ko ek "Reference Book" (PDF ya Database) deta hai.

Technical Flow:

Vector Database: Aapka data chhote-chhote numbers (vectors) mein convert hota hai.

Retrieval: Jab aap sawal puchte hain, AI database mein jaakar relevant "Paragraphs" dhundta hai.

Augmentation: AI un paragraphs ko "Context" ki tarah use karke answer generate karta hai.

Module 4 – Data & Automation for Professionals

RAG

Role: Act as an Academic Analyst and Study Partner.

Task: Review the "LECTURE NOTES" provided below and identify the most critical information.

Guidelines:

Based ONLY on the provided notes, extract the most important data, key concepts, and core takeaways.

Highlight any specific formulas, dates, or terms that are emphasized in the text.

If there are mentions of "Exam-specific" topics or "Important" points by the professor, list them separately.

Keep the summary concise and structured using bullet points for better readability.

LECTURE NOTES: [Paste your notes here]

QUESTION: "Based on the content, what is the most important data and key information from the lectures I have provided?"



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